

ROI ET AGROMET, Thailand

WMO: 484040

Lat: 16.0750N

Lon: 103.6044E

Elev: 156

StdP: 99.47

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.1	15.6	9.1	7.3	17.1	10.6	8.1	18.0	8.2	23.2	7.2	23.2	1.8	40	0.338

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.7	37.6	25.7	36.4	25.8	35.3	25.8	28.0	32.4	27.6	32.0	27.3	31.6	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	23.2	29.7	26.6	22.6	29.5	26.2	22.1	29.2	90.7	32.4	89.1	32.2	87.7	31.8	34.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.7	6.0	5.3	DB	11.3	38.8	1.3	1.3	10.4	39.7	9.7	40.5	9.0	41.2	8.0	42.2	(4)
			WB	9.7	29.5	1.5	1.6	8.6	30.6	7.8	31.5	6.9	32.4	5.8	33.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	23.6	25.8	28.4	30.1	29.7	29.4	28.5	28.2	28.0	27.4	26.2	23.8
	DBStd	2.93	2.65	3.05	2.91	2.32	1.61	1.33	1.53	1.17	1.16	1.28	1.97	2.65
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	1	1	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6359	422	442	571	602	610	582	574	565	539	539	486	428
	CDD18.3	3319	165	208	312	352	352	332	316	307	289	281	236	171
	CDH23.3	36196	1405	2134	3664	4426	4288	4065	3550	3367	3017	2742	2121	1416
	CDH26.7	15353	499	982	1866	2359	2052	1808	1367	1182	977	952	805	506
Wind	WSAvg	1.8	2.1	1.7	1.7	1.5	1.5	1.9	2.0	1.8	1.2	1.6	2.1	2.4
Precipitation	PrecAvg	1461	4	19	32	88	209	198	219	272	308	98	14	2
	PrecMax	1866	41	100	134	180	483	296	407	457	581	224	64	25
	PrecMin	1044	0	0	0	10	56	90	96	48	122	19	0	0
	PrecStd	223	10	29	36	51	111	60	83	122	122	58	20	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	36.4	38.7	39.5	37.6	36.3	36.1	33.7	33.2	33.3	34.1	33.8
		MCWB	24.0	24.1	24.1	25.2	26.7	27.1	25.7	26.9	27.2	25.7	25.9	24.1
	2%	DB	32.2	35.0	37.2	38.0	36.4	34.9	34.2	32.7	32.4	32.3	32.8	32.5
		MCWB	23.6	24.0	24.6	26.1	26.6	26.7	26.4	26.8	26.9	25.5	25.4	23.5
	5%	DB	31.0	33.7	35.8	36.6	35.1	33.8	32.8	31.9	31.7	31.7	32.0	31.1
		MCWB	23.0	23.9	25.0	26.2	26.7	26.6	26.4	26.6	26.7	25.4	24.8	23.3
	10%	DB	29.6	32.2	34.4	35.1	33.8	32.9	31.7	31.1	30.9	30.9	30.9	29.6
		MCWB	22.1	23.8	25.0	26.0	26.7	26.5	26.4	26.5	26.5	25.2	24.2	22.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.3	27.2	28.2	28.4	28.3	27.7	28.0	28.2	27.3	27.0	25.4
		MCDB	31.3	32.6	34.0	34.4	33.2	33.1	31.7	31.3	31.1	30.4	31.9	31.7
	2%	WB	25.1	25.7	26.5	27.5	27.7	27.7	27.2	27.6	27.7	26.9	26.4	24.6
		MCDB	29.6	32.0	33.4	34.1	32.8	32.5	31.2	30.9	30.8	29.9	31.1	30.6
	5%	WB	24.2	25.1	25.9	26.9	27.3	27.2	27.0	27.2	27.3	26.4	25.8	23.9
		MCDB	28.5	31.4	33.0	33.7	32.7	31.9	31.0	30.5	30.3	29.5	30.1	29.5
	10%	WB	23.1	24.4	25.4	26.5	27.0	26.9	26.7	26.9	27.0	26.0	25.2	23.2
		MCDB	28.1	30.4	32.3	33.0	32.5	31.4	30.4	30.1	29.9	29.1	29.1	28.6
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.0	10.8	9.7	8.1	6.8	6.0	5.7	5.7	7.0	9.1	11.0
		MCDBR	12.4	13.0	12.3	11.1	9.7	8.1	7.8	6.7	6.8	7.9	9.9	12.0
		MCWBR	4.7	4.2	3.4	3.2	2.8	2.3	2.4	2.5	2.7	2.9	3.4	4.2
	5% WB	MCDBR	10.4	11.0	10.8	10.1	8.7	7.4	6.7	6.3	6.4	7.1	8.9	11.0
	MCWBR	4.4	3.9	3.6	3.4	3.1	2.6	2.5	2.6	2.8	3.0	3.3	4.6	
Clear-Sky Solar Irradiance	taub	0.476	0.593	0.661	0.619	0.513	0.469	0.459	0.463	0.506	0.525	0.471	0.465	
	taud	2.085	1.791	1.682	1.818	2.139	2.281	2.311	2.289	2.157	2.071	2.196	2.146	
	Ebn at Noon	809	731	696	728	798	826	836	840	807	779	810	807	
	Edh at Noon	156	219	251	220	157	135	131	135	154	163	139	144	
All-Sky Solar Radiation	RadAvg	4.93	5.20	5.30	5.71	5.57	5.21	4.89	4.76	4.64	5.04	4.94	4.77	
	RadStd	0.28	0.25	0.23	0.25	0.30	0.25	0.35	0.31	0.34	0.34	0.18	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.27	N/A	N/A	N/A	+0.15	+0.13	N/A	N/A	N/A	N/A

Nomenclature: See separate page